Chemistry of the Earth: Lecture 5

Chemical Reactions and Oxidation State



Part 1: Balancing chemical equations

OBJECTIVES OF THIS LECTURE AND PRACTICAL

- Review how chemical formulae and chemical reactions are written, and how to balance a chemical equation

- Review the meaning of moles in chemistry

- Understand the concept of oxidation state, and variable oxidation states among the transition elements

- Investigate some important oxidation-reduction reactions in geochemical systems







READING AND RESOURCES

(AMD: Acid Mine Drainage BIF: Banded Iron Formation)

Grotzinger & Jordan (Basic: Minimum reading)	**BIF - p.682-684; AMD - p. 674**
Misra (Recommended)	**Moles – p. 14; Redox - Chapter 8.1 (p. 167-168)**
Gill (Recommended)	**BIF - Chapter 12 (p.222-224)**
White	Redox - Chapter 3.11 (p. 100)
McSween & Richardson	Redox – p.124; U – p. 132-133; **BIF – p.164-166**

Internet resources

AMD: Environment Agency: Abandoned Mines and the Water Environment
https://www.gov.uk/government/publications/abandoned-mines-and-the-water-environment (also posted as a pdf on BB)

BIF: https://www.coursera.org/lecture/earth-amnh/banded-iron-formation-tAQ0b

BIF: https://www.youtube.com/watch?v=Y6haZqkfy_w

CHEMICAL FORMULAE

The chemical formula of a compound tells you how many atoms of each element the molecule contains. (GCSE definition.)

The chemical formula of a mineral gives the atomic proportions of each element in the mineral.

A chemical formula must have a zero charge overall.

Mineral	Charges on ions	Formula
Water	H^+ O^{2-}	H_2O
Halite (salt)	Na^+ Cl^-	$NaCl$
Rutile	Ti^{4+} O^{2-}	TiO_2
Corundum	Al^{3+} O^{2-}	Al_2O_3
Calcite	Ca^{2+} $(CO_3)^{2-}$	$CaCO_3$
Forsterite	Mg^{2+} $(SiO_4)^{4-}$ (Si^{4+} and O^{2-})	Mg_2SiO_4

Practical 2

• Brackets around a complex ion show that there are multiple units of that ion:

Brucite	Mg^{2+} $(OH)^-$	$Mg(OH)_2$
Fluorapatite	Ca^{2+} $(PO_4)^{3-}$ F^-	$Ca_5(PO_4)_3F$

Element **substitutions** in minerals

Mineral	Charges on ions	Formula
Forsterite	$Mg^{2+}\ (SiO_4)^{4-}$	Mg_2SiO_4
Fayalite	$Fe^{2+}\ (SiO_4)^{4-}$	Fe_2SiO_4

OLIVINE is a solid solution between forsterite and fayalite,
i.e. a crystal of olivine can have any ratio of Mg/Fe atoms:

Olivine	$Mg^{2+}\ \ Fe^{2+}\ (SiO_4)^{4-}$	$(Mg,Fe)_2SiO_4$



wikipedia

Albite	$Na^{+}\ \ Al^{3+}\ (Si_3O_8)^{4-}$	$NaAlSi_3O_8$
Orthoclase	$K^{+}\ \ Al^{3+}\ (Si_3O_8)^{4-}$	$KAlSi_3O_8$

ALKALI FELDSPAR is a solid solution between albite and orthoclase,
i.e. a crystal of alkali feldspar can have any ratio of Na/K atoms:

Alkali Feldspar	$Na^{+}\ K^{+}\ \ Al^{3+}\ (Si_3O_8)^{4-}$	$(Na,K)AlSi_3O_8$



pitt.edu

- For element substitutions, atoms in brackets can be exchanged.
The exchangeable atoms are separated by a comma.

CHEMICAL EQUATION

A chemical equation describes a reaction or a chemical equilibrium:

$A + B \rightarrow C + D$ (reaction) $\qquad$ $A + B = C + D$ (equilibrium)

The two sides of the reaction must

- have the same number of atoms of each element
- have the same charge

$Mg + Cl_2 \rightarrow MgCl_2$

$2Na + Cl_2 \rightarrow 2NaCl$

Balance these equations:

$Na + O_2 \rightarrow Na_2O$

$Na + H_2O \rightarrow NaOH + H_2$

CHEMICAL EQUATION

A chemical equation describes a reaction or a chemical equilibrium:

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The two sides of the reaction must have the same number of atoms of each element, and must have the same charge.

$Mg + Cl_2 \rightarrow MgCl_2$

$2Na + Cl_2 \rightarrow 2NaCl$

Balanced equations:

$4Na + O_2 \rightarrow 2Na_2O$

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

GCSE level reactions (http://www.bbc.co.uk/schools/gcsebitesize)

Write the balanced equation for the reaction between calcium and water to produce calcium hydroxide and hydrogen.

Step 1

Write the formulae for each substance:

$Ca + H_2O \rightarrow Ca(OH)_2 + H_2$

GCSE level reactions (http://www.bbc.co.uk/schools/gcsebitesize)

Write the balanced equation for the reaction between calcium and water to produce calcium hydroxide and hydrogen.

Step 1

Write the formulae for each substance:

$Ca + H_2O \rightarrow Ca(OH)_2 + H_2$

Step 2

Check that each element is balanced.

There is one Ca atom on each side – OK, Ca is balanced.

Check for one unbalanced element, for example, O. Adjust the number of each atom or molecule needed, but never change a formula. We need two O atoms on each side:

$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$

Check the remaining element, H. There are 4 H atoms on each side – OK, equation is now balanced.

REACTIONS INVOLVING MINERALS

Some important reactions involving minerals include:

- response to changes in pressure and temperature in metamorphic rocks
- reactions of minerals with water (hydration) and other fluids (e.g. carbonation)
- loss of volatile species from minerals (dehydration, decarbonation)
- reactions between minerals and water during weathering
- precipitation of minerals from water and other fluids

Example 1: Response to change in P and T in metamorphic rocks. The reaction of albite to jadeite + quartz occurs in subduction zones, with increase in pressure.

Reaction: albite → jadeite + quartz

$NaAlSi_3O_8$ → $NaAlSi_2O_6$ + SiO_2

albite → jadeite + quartz

Na	1	1	0
Al	1	1	0
Si	3	2	1
O	8	6	2

The reaction is balanced as written.

Example 2: Reaction of minerals with water. Around a divergent plate boundary on the ocean floor, olivine in basalt and gabbro reacts with water to form serpentine and brucite. This process is called serpentinization.

Reaction:

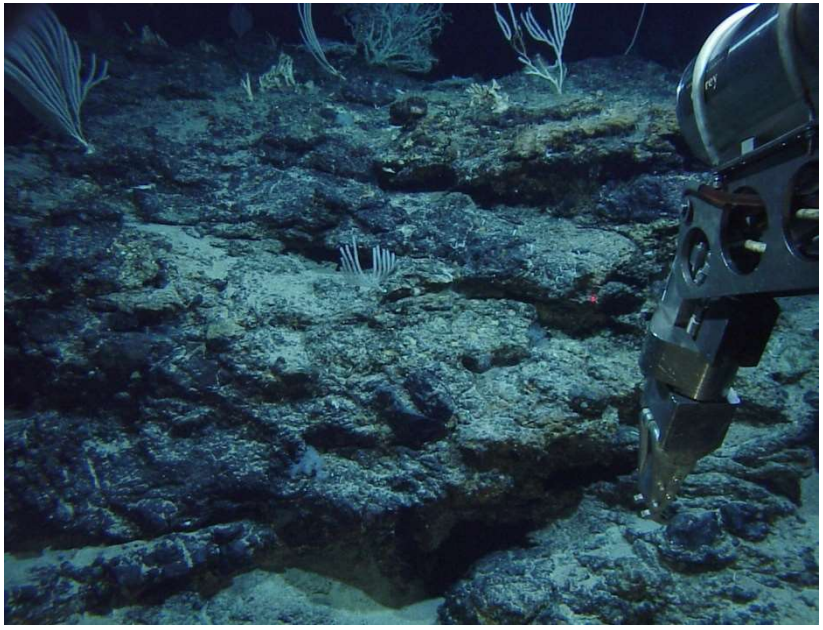
Olivine + Water → Serpentine + Brucite

$Mg_2SiO_4 + H_2O \rightarrow Mg_3Si_2O_5(OH)_4 + Mg(OH)_2$

(not balanced)



Serpentinite: Golden Gate National Recreation Area, California, USA. Eurico Zimbres



Serpentinized peridotite.
NOAA: Lost City, Mid-Atlantic Ridge

Reaction:

Olivine + Water → Serpentine + Brucite

Mg_2SiO_4 + H_2O → $Mg_3Si_2O_5(OH)_4$ + $Mg(OH)_2$



Serpentinite

Examine the reaction:

Element	Olivine	Water	Total LHS	=	Serpentine	Brucite	Total RHS
Mg	2	0	2				
Si	1	0	1				
O	4	1	5				
H	0	2	2				

Reaction:

Olivine + Water → Serpentine + Brucite

Mg_2SiO_4 + H_2O → $Mg_3Si_2O_5(OH)_4$ + $Mg(OH)_2$



Serpentinite

Element	Olivine	Water	Total LHS	=	Serpentine	Brucite	Total RHS
Mg	2	0	2		3	1	4
Si	1	0	1		2	0	2
O	4	1	5		9	2	11
H	0	2	2		4	2	6

- Not balanced! – none of the elements are balanced.
- Start with Si, since it only appears in one formula on each side.
- Si = 1 on LHS, Si = 2 on RHS. Multiply Olivine by 2.

Reaction:

Olivine + Water → Serpentine + Brucite

2 Mg_2SiO_4 + H_2O → $Mg_3Si_2O_5(OH)_4$ + $Mg(OH)_2$



Serpentinite

Element	Olivine **X2**	Water	Total LHS	=	Serpentine	Brucite	Total RHS
Mg	2 **/ 4**	0	2 **/ 4**		3	1	4
Si	1 **/ 2**	0	1 **/ 2**		2	0	2
O	4 **/ 8**	1	5 **/ 9**		9	2	11
H	0	2	2		4	2	6

- Mg and Si are now balanced.
- O and H are not balanced. Balance H next, since it only appears in one formula (water) on LHS. As a general rule, leave O until last.
- H = 2 on LHS, H = 6 on RHS. Multiply water by 3.

Reaction:

Olivine + Water → Serpentine + Brucite

2 Mg_2SiO_4 + **3** H_2O → $Mg_3Si_2O_5(OH)_4$ + $Mg(OH)_2$



Serpentinite

Element	Olivine **X2**	Water **X3**	Total LHS	=	Serpentine	Brucite	Total RHS
Mg	2 **/ 4**	0	2 **/ 4**		3	1	4
Si	1 **/ 2**	0	1 **/ 2**		2	0	2
O	4 **/ 8**	1 **/ 3**	5 **/ 9 / 11**		9	2	11
H	0	2 **/ 6**	2 **/ 6**		4	2	6

The reaction is now balanced for all elements

Example 3: Breakdown of hydrated minerals releasing H_2O.
Many sedimentary rocks contain water-bearing minerals that break down and release water (dehydration reactions). In low-grade metamorphism, pyrophyllite breaks down to form kyanite, quartz and water.

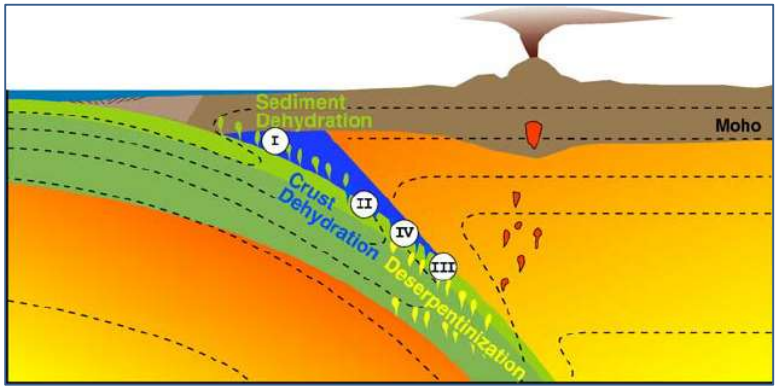
Pyrophyllite <=> Kyanite + Quartz + Water

$Al_2Si_4O_{10}(OH)_2$ <=> Al_2SiO_5 + SiO_2 + H_2O

Balance the reaction.

Element	Pyrophyllite	Total LHS	=	Kyanite	Quartz	Water	Total RHS
Al	2	2		2			2
Si	4	4		1	1		2
O	12	12		5	2	1	8
H	2	2				2	2

- Al and H are balanced.
- As a general rule, leave O until last.
- So, balance Si: multiply Quartz by 3.

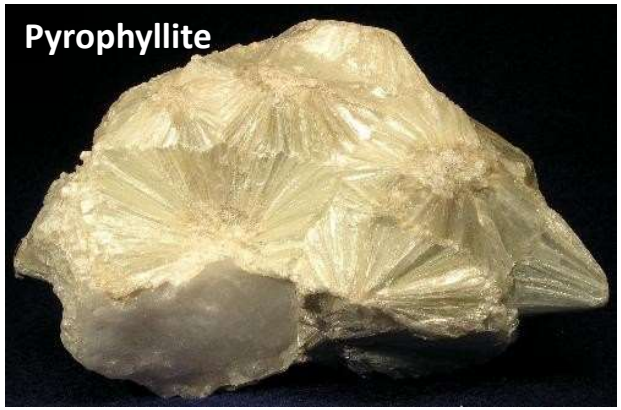


Pyrophyllite <=> Kyanite + Quartz + Water

$Al_2Si_4O_{10}(OH)_2$ <=> Al_2SiO_5 + **3**SiO_2 + H_2O

Element	Pyrophyllite	Total LHS	=	Kyanite	Quartz **X3**	Water	Total RHS
Al	2	2		2			2
Si	4	4		1	1 **/ 3**		2 **/ 4**
O	12	12		5	2 **/ 6**	1	8 **/ 12**
H	2	2				2	2

The reaction is now balanced for all elements



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Wikipedia

Chemistry of the Earth: Lecture 5

Chemical Reactions and Oxidation State



Part 2: Moles and oxidation state

MOLES AND THE AVOGADRO CONSTANT

MOLE: SI unit for "amount of substance". Unit is written as **mol**.

One MOLE is the amount of a chemical substance that contains as many entities (atoms, molecules, ions, etc.) as there are atoms in 12 g of carbon-12 (^{12}C).



Wikipedia

Lorenzo Romano Amedeo Carlo Avogadro di Quaregna e di Cerreto, 1776-1856

Question:
How many atoms are there in 12 g of ^{12}C?

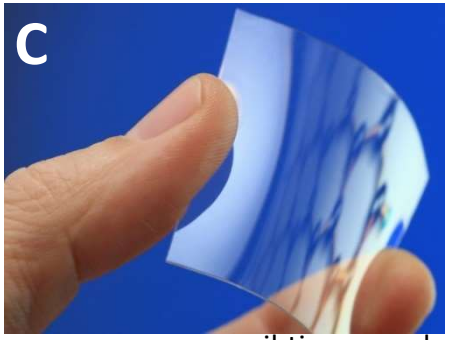
Answer:
6.022×10^{23}

This number is the **Avogadro constant.**

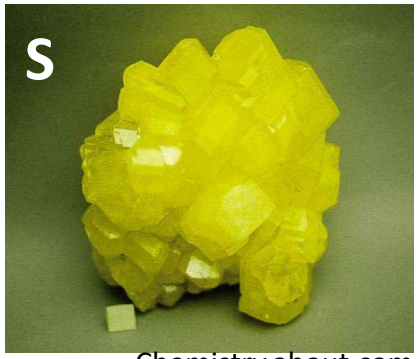
The **Avogadro constant** is the number of constituent particles, usually atoms or molecules, that are contained in one mole of the substance. (Number of molecules per mole.)

The **Avogadro constant** is defined such that the mass of one mole of a substance, expressed in grams, is equal to the atomic or molecular mass of the substance.

12 g of ^{12}C has 6.022×10^{23} carbon atoms
1 g of ^{1}H atoms has 6.022×10^{23} hydrogen atoms
32 g of ^{32}S atoms has 6.022×10^{23} sulphur atoms
.....etc.....



ibtimes.co.uk

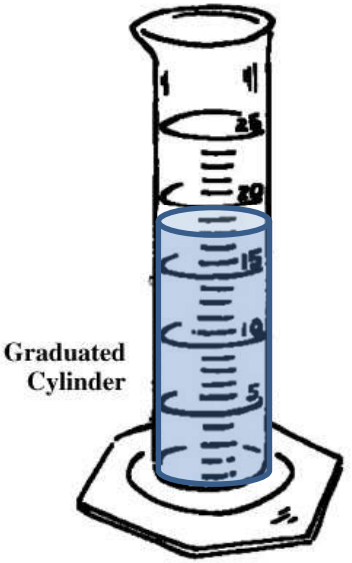


Chemistry.about.com

Water: H_2O

Atomic mass of H is 1
Atomic mass of O is 16

- One molecule of H_2O has a **molecular mass** of 18
- One mole of water has a mass of 18 g
- 18 g of water has 6.022×10^{23} water molecules



Molecular mass (or **molecular weight**) is the mass of a molecule. It is calculated as the sum of the atomic mass of each constituent atom multiplied by the number of atoms of that element in the molecular formula.

What is the molecular mass of quartz?
Atomic mass of Si = 28.086
Atomic mass of O = 15.999



Wikipedia

Approximately how many silicon atoms are there in a quartz crystal that weighs 120 g ? How many oxygen atoms?

Molecular mass (or **molecular weight**) is the mass of a molecule. It is calculated as the sum of the atomic mass of each constituent atom multiplied by the number of atoms of that element in the molecular formula.

What is the molecular mass of quartz?
Atomic mass of Si = 28.086
Atomic mass of O = 15.999

Formula: SiO_2
Molecular mass:
28.086 + (2 * 15.999) = 60.084



Wikipedia

Approximately how many silicon atoms are there in a quartz crystal that weighs 120 g ? How many oxygen atoms?

Si atoms: ~12×10^{23}
O atoms: ~24×10^{23}

Each dish contains one mole of ionic compound

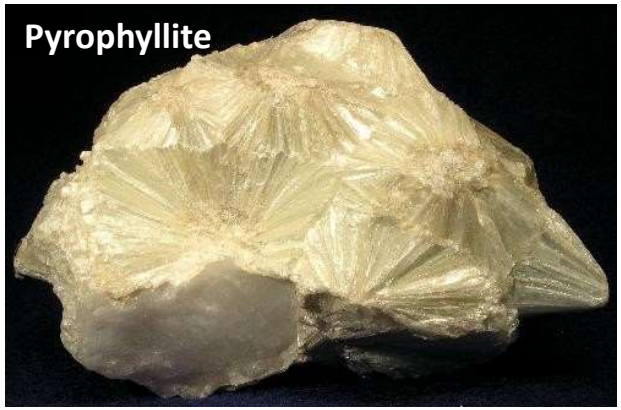


www.rsc.org

Example 3 above: Breakdown of hydrated minerals releasing H_2O.

Pyrophyllite	<=>	Kyanite or Andalusite	+	Quartz	+	Water
$Al_2Si_4O_{10}(OH)_2$	<=>	Al_2SiO_5	+	$3SiO_2$	+	H_2O

"One mole of pyrophyllite breaks down to one mole of Al_2SiO_5, 3 moles of quartz and one mole of water."







Wikipedia

OXIDATION – REDUCTION REACTIONS (REDOX REACTIONS)

Lecture 2

Some reactions involve a change in the VALENCE of an element.

Valence – charge on an ion.
Example: Fe^{2+} (divalent iron ion), Fe^{3+} (trivalent iron ion)

Hematite, Fe_2O_3 -- $Fe^{3+}{}_2O^{2-}{}_3$ Magnetite, Fe_3O_4 -- $Fe^{2+}Fe^{3+}{}_2O^{2-}{}_4$

Hematite = Magnetite + O_2
$6\ Fe_2O_3$ = $4\ Fe_3O_4$ + O_2

Valence can also be described in terms of OXIDATION STATE.

Oxidation state of iron is Fe^{2+} (ferrous), Fe^{3+} (ferric).

The above reaction involves a *change of oxidation state*, so it is described as an oxidation-reduction reaction or a REDOX reaction.

OXIDATION – REDUCTION REACTIONS (REDOX REACTIONS)

Hematite = Magnetite + O_2

$6\ Fe^{3+}{}_2O^{2-}{}_3 = 4\ Fe^{2+}Fe^{3+}{}_2O^{2-}{}_4 + O_2$





- An Fe^{3+} ion in hematite gains an electron to form Fe^{2+}, and is therefore reduced. Reduction = decrease in oxidation state.

  $Fe^{3+} + e^- = Fe^{2+}$ (reduction of Fe^{3+})

- In the reverse reaction, the Fe^{2+} ion in magnetite loses an electron to form Fe^{3+}, and is therefore oxidized. Oxidation = increase in oxidation state.

  $Fe^{2+} = Fe^{3+} + e^-$ (oxidation of Fe^{2+})

REDOX reactions do not necessarily need to involve oxygen.

Oxidation corresponds to LOSS of an electron ($Fe^{2+} \rightarrow Fe^{3+}$)
Reduction corresponds to GAIN of an electron ($Fe^{3+} \rightarrow Fe^{2+}$)

Example 1: This reaction shows how zinc metal can react with copper ions in a solution (aq) to form copper metal and dissolved zinc ions:

$$Zn_s + Cu^{2+}_{aq} = Zn^{2+}_{aq} + Cu_s$$

Zn is oxidized to Zn^{2+} ions (loss of electrons): $Zn = Zn^{2+} + 2e^-$

Cu ions are reduced to Cu (gain of electrons): $Cu^{2+} + 2e^- = Cu$

Half reactions

Example 2: Reaction of chlorine gas with metallic iron:

$$2\ Fe + 3\ Cl_2 = 2\ Fe^{3+} + 6\ Cl^-$$

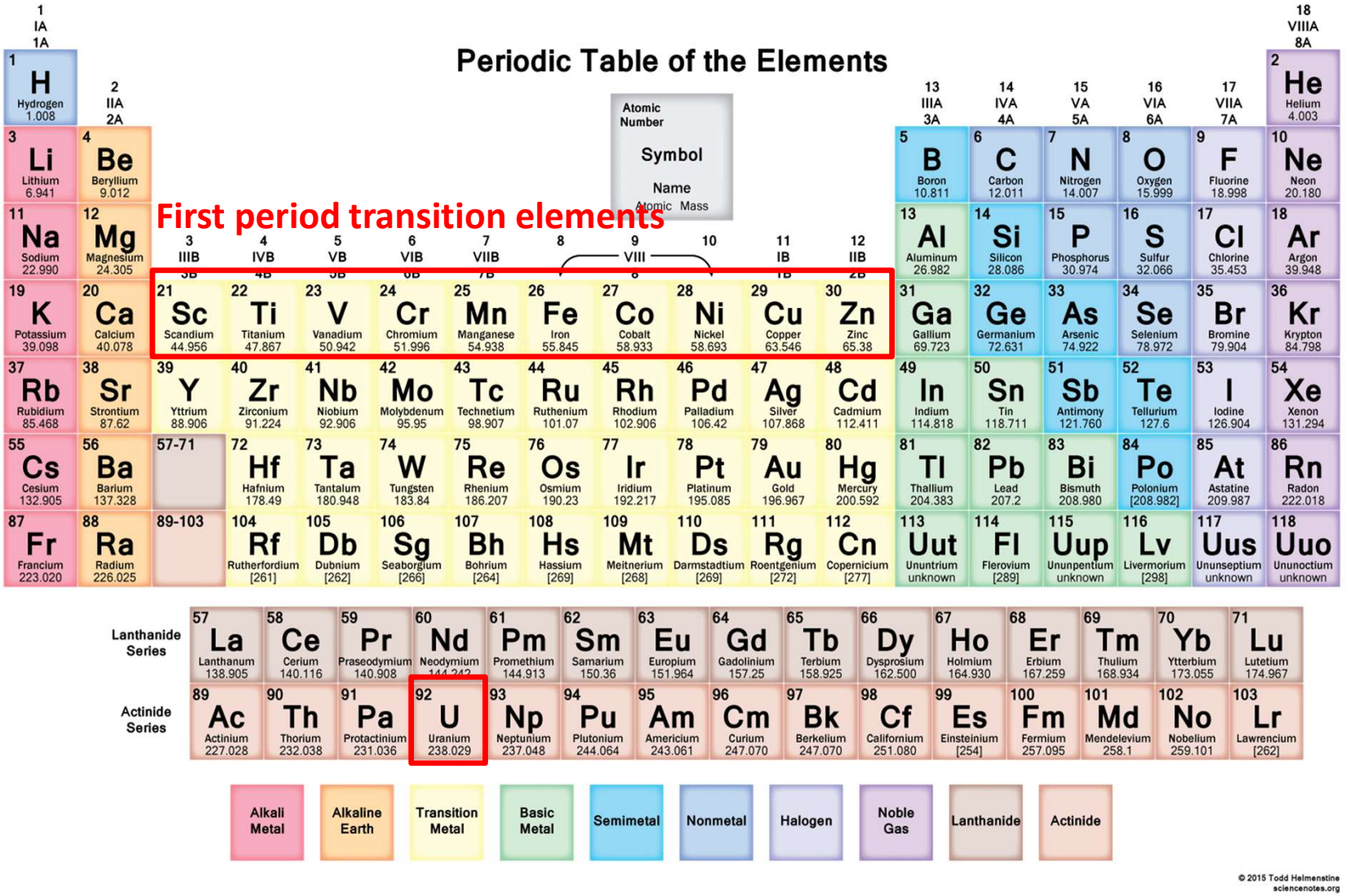
Fe^0 is oxidized to Fe^{3+} (loss of electrons): $2\ Fe = 2\ Fe^{3+} + 6\ e^-$

Cl^0_2 is reduced to Cl^- (gain of electrons): $3\ Cl_2 + 6\ e^- = 6\ Cl^-$

Half reactions

Note that the charge must balance in redox reactions, as well as the atoms

VARIABLE OXIDATION STATE: TRANSITION METALS



Transition elements – filling the d orbitals.

Electron configurations can vary.

Oxidation states of first-series transition metals: (most common ions highlighted)

21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
								1+	
		2+	2+	2+	2+	2+	2+	2+	2+
3+	3+	3+	3+	3+	3+	3+	3+	3+	
	4+	4+	4+	4+					
		5+							
			6+	6+	6+				
				7+					

Cnx.org

Also oxidation states of 0: Fe^0 = iron metal

Uranium (U): Another element in which oxidation state is important

- U^{6+} (more oxidized) is soluble in water
- U^{4+} (more reduced) is much less soluble

Uranium deposits are formed when an oxidized solution is reduced.



Uraninite (UO_2) crystals (also known as pitchblende) from Topsham, Maine (size: 2.7×2.4×1.4 cm).
Uraninite is the most common ore mined to extract uranium.

Wikipedia

Chemistry of the Earth: Practical 5

Chemical Reactions and Oxidation State



Practical 5:

Chemical reactions and oxidation state

- Practice balancing chemical reactions relevant to minerals and rocks
- Practice calculations involving moles and molecular mass
- Investigate some important oxidation-reduction reactions in geochemical systems

Example 3: Breakdown of hydrated minerals releasing H_2O.
Many sedimentary rocks contain water-bearing minerals that break down and release water (dehydration reactions). In low-grade metamorphism, pyrophyllite breaks down to form kyanite, quartz and water.

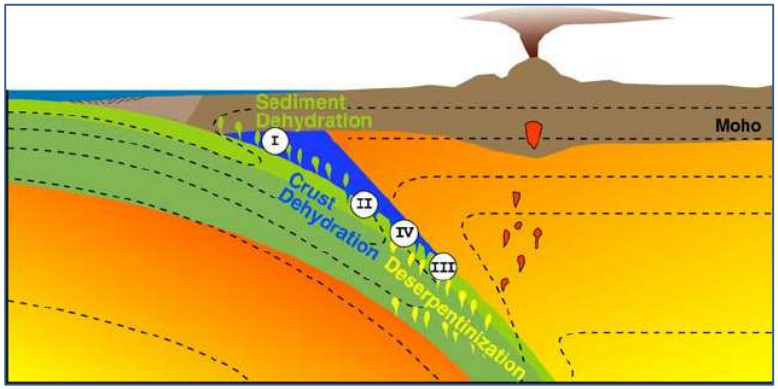
Pyrophyllite <=> Kyanite + Quartz + Water

$Al_2Si_4O_{10}(OH)_2$ <=> Al_2SiO_5 + SiO_2 + H_2O

Balance the reaction.

Element	Pyrophyllite	Total LHS	=	Kyanite	Quartz	Water	Total RHS
Al	2	2		2			2
Si	4	4		1	1		2
O	12	12		5	2	1	8
H	2	2				2	2

- Al and H are balanced.
- As a general rule, leave O until last.
- So, balance Si: multiply Quartz by 3.

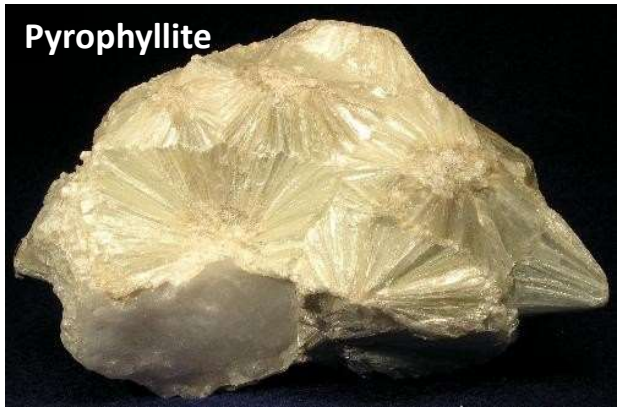


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Si	4	4		1	1 **/ 3**		2 **/ 4**
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H	2	2				2	2

The reaction is now balanced for all elements



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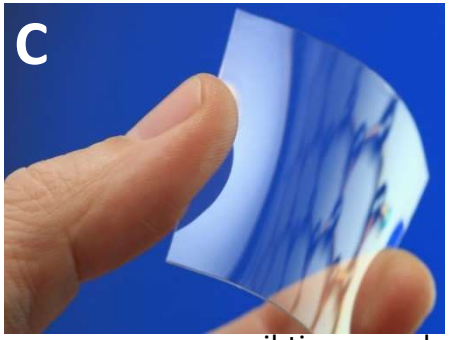




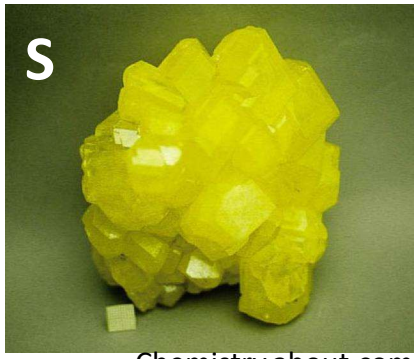
Wikipedia

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ibtimes.co.uk

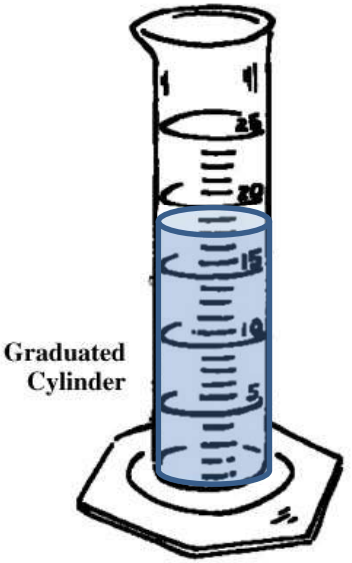


Chemistry.about.com

Water: H_2O

Atomic mass of H is 1
Atomic mass of O is 16

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- One mole of water has a mass of 18 g
- 18 g of water has 6.022×10^{23} water molecules

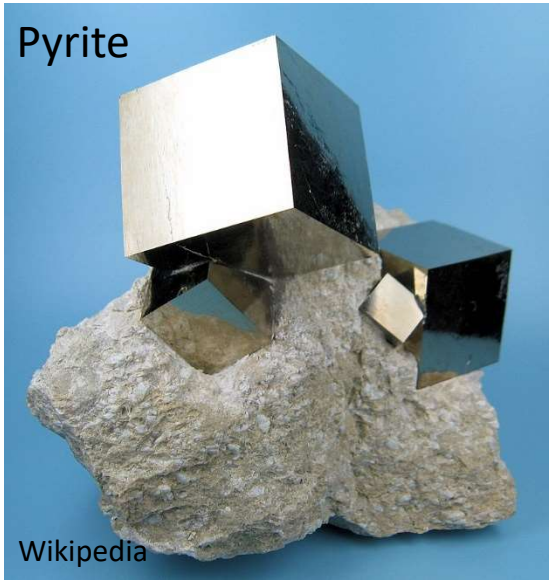


Minerals involved in oxidation-reduction reactions

Uraninite



Uraninite (UO_2) crystals (also known as pitchblende) from Topsham, Maine (size: 2.7×2.4×1.4 cm).



Pyrite (FeS_2) cubic crystals on marlstone, from Navajún, Rioja, Spain. Size: 95 mm x 78 mm. Main crystal: 31 mm on edge. Mass: 512 g.

Acid mine drainage



Iron hydroxide precipitate in a Missouri stream receiving acid drainage from surface coal mining



Rio Tinto, Spain

Banded Iron Formation (BIF)



Banded Iron Formation, Karijini National Park, Western Australia

Wikipedia: https://www.flickr.com/people/30659367@N00